- ***CSD – SQL Databases***
- ***Group Assignment: UML Class Diagram Documentation***
- ***Kigali Independent University (ULK)***

• Introduction

This document presents the structural design of the Event & Conference Management System using a UML Class Diagram. It outlines the main classes, their attributes, operations, and the relationships connecting them.

2. Detailed Class Specifications

1. Attendee Class

Attributes:

attendeeID (Integer, Primary Key)

name (String)

email (String)

phone (String)

Methods / Operations:

registerEvent( )

updateProfile( )

➢ Speaker Class

Attributes:

speakerID (Integer, Primary Key)

name (String)

expertise (String)

bio (String)

Methods / Operations:

assignSession( )

viewSchedule( )

➢ Session Class

Attributes:

sessionID (Integer, Primary Key)

title (String)

time (DateTime)

location (String)

Methods / Operations:

addSession( )

updateDetails( )

➢ Ticket Class

Attributes:

ticketID (Integer, Primary Key)

type (String)

price (Float)

status (String)

Methods / Operations:

bookTicket( )

cancelTicket( )

➢ Relationships & Multiplicity

Attendee to Ticket: One Attendee can book one or more Tickets (1 to 0..*), and each Ticket belongs to a specific Attendee (1 to 1).

Speaker to Session: One Speaker can present one or more Sessions (1 to 0..*), and each Session is delivered by a Speaker (1 to 1).

Attendee to Session: Attendees can register to attend multiple Sessions (Many-to-Many relationship handled through registration).